Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is 0.333… (the 3 repeats forever) a rational number? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 8 · 8.NS.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is $\sqrt{2}$ a rational number? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is 0.25 a rational number? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Between which two whole numbers is $\sqrt{50}$? Student's answer: between 5 and 6

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Locate $\sqrt{10}$ on a number line. Student says: It is closer to 4.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Compare $\sqrt{7}$ and 2.6. Student says: $\sqrt{7}$ is less than 2.6.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 8 · 8.EE.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Simplify $2^3 \times 2^2$. Student's answer: 2^6

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Evaluate 5^0. Student's answer: 0

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Evaluate 10^{-2}. Student's answer: -100

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find $\sqrt{49}$. Student's answer: 24.5

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $x^2 = 36$ for the positive value of x. Student's answer: $x = 3$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 8 · 8.EE.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find $\sqrt[3]{27}$. Student's answer: 9

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 8 · 8.EE.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Write 4.5×10^3 in standard form. Student's answer: 450

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Write 0.00052 in scientific notation. Student writes: 5.2×10^4

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Which is greater: 3×10^4 or 5×10^3? Student's answer: 5×10^3

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A city has 2.4×10^5 people. How many people are in 3 cities of the same size? Student's answer: 7.2×10^6

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Divide $(8 \times 10^6) \div (2 \times 10^2)$. Student's answer: 4×10^3

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 8 · 8.EE.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Multiply $(6 \times 10^{-3})(5 \times 10^{-2})$. Student's answer: 3×10^{-5}

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A line goes through (0, 0) and (2, 6). What is its slope? Student's answer: $\frac{1}{3}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 8 · 8.EE.5

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A line rises 4 units for every 2 units across. What is its slope? Student's answer: $\frac{1}{2}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 8 · 8.EE.5

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A line goes through (1, 5) and (3, 1). What is its slope? Student's answer: 2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A line goes through (0, 0) and (1, 4). Write its equation. Student writes: $y = x + 4$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 8 · 8.EE.6

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A line has a slope of 2 and a y-intercept of 3. Write its equation. Student writes: $y = 3x + 2$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A line goes through (0, -2) and (2, 2). Write its equation. Student writes: $y = -2x + 2$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 8 · 8.EE.7

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $3(x + 2) = 2x + 9$. Student's answer: $x = 15$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $2x + 3 = 2x + 5$. Student's answer: $x = 1$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $4(x + 1) = 4x + 4$. Student's answer: $x = 0$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 8 · 8.EE.8

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve the system $y = x + 1$ and $y = 3x - 3$. Student's answer: $(1, 2)$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve the system $x + y = 10$ and $x - y = 4$. Student's answer: $(3, 7)$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 8 · 8.EE.8

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve the system $y = 2x + 1$ and $y = 2x + 5$. Student's answer: $(1, 3)$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is the relation {(1, 2), (1, 3), (2, 4)} a function? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is y = 2x a function? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 8 · 8.F.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A graph crosses a vertical line in two places. Is the graph a function? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Function A has a rate of change of 3 from a table. Function B is $y = 2x + 3$. Which has the greater rate of change? Student's answer: Function B

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Function A is y = 3x + 1. Function B has the points (0, 2) and (1, 5). Which has the greater starting value? Student's answer: Function A

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Function A is y = 5x. Function B has the points (0, 0) and (1, 4). Which has the greater rate of change? Student's answer: Function B

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find y when $y = x^2$ and $x = 3$. Student's answer: 6

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Which function is nonlinear: $y = 4x + 1$ or $y = x^2$? Student's answer: $y = 4x + 1$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

The table has x = 1, 2, 3 and y = 1, 4, 9. Is the relationship linear? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 8 · 8.F.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A table has the points (0, 3) and (2, 7). Write the linear function. Student writes: $y = 3x + 2$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the initial value of y = 5x - 4? Student's answer: 5

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 8 · 8.F.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A tank has 10 gallons and loses 2 gallons each hour. Write the function. Student's answer: starting value 10 and rate 10

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A graph starts high on the left and falls steadily to the right. Is it increasing or decreasing?

Student's answer: increasing

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A graph is flat for a while and then rises steadily. Which description fits the flat part? Student's answer: increasing at a constant rate

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 8 · 8.F.5

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A temperature graph rises from 40 to 80 and then falls to 50. When is the temperature decreasing? Student's answer: from 40 to 80

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Translate the point (2, 3) right 4 units. Student's answer: (2, 7)

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Reflect the point (2, 3) across the x-axis. Student's answer: (-2, 3)

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 8 · 8.G.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Reflect the point (2, 3) across the y-axis. Student's answer: (2, -3)

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Are two rectangles that are both 3 by 5 congruent, with one rotated? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Are triangles with sides 3, 4, 5 and 4, 5, 6 congruent? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A triangle is reflected across a line to make a new triangle. Are they congruent? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Dilate the point (3, -1) by a scale factor of 2 from the origin. Student's answer: (5, 1)

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Dilate the point (6, 10) by a scale factor of $\frac{1}{2}$ from the origin. Student's answer: (3, 4)

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A shape has a side of 2 units. It is dilated by a scale factor of 3. What is the new side length?

Student's answer: 5 units

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Triangles have sides 3, 4, 5 and 6, 8, 10. Are they similar? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 8 · 8.G.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A side of 4 in one triangle matches a side of 6 in a similar triangle. The matching side is 10 in the first triangle. What is the matching side in the second? Student's answer: 12

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Rectangles are 2 by 3 and 4 by 6. Are they similar? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A triangle has angles 50° and 60°. What is the third angle? Student's answer: 80°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 8 · 8.G.5

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Two parallel lines are cut by a transversal. One alternate interior angle is 65°. What is the other?

Student's answer: 115°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Parallel lines are cut by a transversal. A corresponding angle is 40°. What is the matching corresponding angle? Student's answer: 140°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 8 · 8.G.6

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A right triangle has legs 3 and 4. What is the hypotenuse? Student's answer: 7

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 8 · 8.G.6

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A right triangle has legs 6 and 8. What is the hypotenuse? Student's answer: 14

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A right triangle has a hypotenuse of 13 and a leg of 5. What is the other leg? Student's answer: 8

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A rectangular floor is 12 feet by 5 feet. What is the length of the diagonal? Student's answer: 17 feet

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A ladder 10 feet long leans on a wall with its base 6 feet from the wall. How high does it reach?

Student's answer: 16 feet

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A box is 3 by 4 by 12 units. What is the length of its space diagonal? Student's answer: 19

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the distance between (0, 0) and (6, 8). Student's answer: 14

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the distance between (1, 1) and (4, 5). Student's answer: 7

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the distance between (−2, 3) and (1, −1). Student's answer: 7

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the volume of a cylinder with radius 2 and height 3. Student's answer: 6π

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the volume of a cone with radius 3 and height 4. Student's answer: 36π

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the volume of a sphere with radius 3. Student's answer: 12π

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.